

SQL Intelligence — PostgreSQL Practice Questions

110 practice questions + 10 final challenges based on the SQL topics and functions you have studied.

Practice database: customers table with id, name, email, age, and create_at. Use the separate SQL database script provided with this practice set.

Recommended rule: Try each question yourself before checking any solution. Write the SQL query, execute it, and inspect the result.

LEVEL 1 — Basic Filtering

- Q1. Display all customers.
- Q2. Display customers whose age is greater than 25.
- Q3. Display customers whose age is less than 30.
- Q4. Display customers whose age is exactly 20.
- Q5. Display customers whose age is greater than 25 and less than 40.
- Q6. Display customers whose age is NOT less than 30.
- Q7. Display customers whose age is below 25 and whose name starts with A.
- Q8. Display customers whose age is above 30 and whose name starts with A.

LEVEL 2 — Sorting + Pagination

- Q9. Display all customers ordered by age from youngest to oldest.
- Q10. Display all customers ordered by age from oldest to youngest.
- Q11. Display customers ordered alphabetically by name.
- Q12. Display customers ordered by name ascending and age ascending.
- Q13. Display the 5 youngest customers.
- Q14. Display the 5 oldest customers.
- Q15. Display the first 5 customers after sorting by age.
- Q16. Display the next 5 customers after skipping the first 5 oldest customers.
- Q17. Display the 5 customers with the highest IDs.

LEVEL 3 — DISTINCT / BETWEEN / IN

- Q18. Display all unique ages.
- Q19. Display customers whose age is between 20 and 30.
- Q20. Display customers whose age is between 18 and 25, ordered by age.
- Q21. Display customers whose age is exactly one of: 18, 20, 25, 30.
- Q22. Display customers whose age is NOT between 20 and 30.
- Q23. Display unique ages between 20 and 35.

LEVEL 4 — ILIKE

- Q24. Find customers whose name starts with A.
- Q25. Find customers whose name starts with K.
- Q26. Find customers whose name ends with a.
- Q27. Find customers whose name contains an.
- Q28. Find customers whose email contains gmail.
- Q29. Find customers whose email ends with .com.
- Q30. Find customers whose name starts with A and age is below 30.

LEVEL 5 — GROUP BY + Aggregation

- Q31. Find the total number of customers.
- Q32. Find the total sum of all customer ages.
- Q33. Find the average customer age.
- Q34. Find the average age rounded to 2 decimal places.
- Q35. Find the youngest customer's age.
- Q36. Find the oldest customer's age.
- Q37. Find the number of customers for each age.
- Q38. Find how many customers belong to each age between 18 and 30.
- Q39. Show only age groups that have more than 1 customer.
- Q40. Show the age and customer count, ordered by customer count from highest to lowest.

LEVEL 6 — CASE

- Q41. Create an Age_Group column: age < 20 = Teen; 20–29 = Young; 30–39 = Adult; 40+ = Senior.
- Q42. Display name, age, and Age_Group.
- Q43. Create this classification: age < 25 = Junior; 25–34 = Middle; 35+ = Senior.
- Q44. Count how many customers belong to each age group.
- Q45. Sort customers according to their age group.

LEVEL 7 — ALTER TABLE

- Q46. Add a phone VARCHAR(15) column to customers.
- Q47. Add an is_active BOOLEAN column with default TRUE.
- Q48. Add a country VARCHAR(20) column with default India and make it NOT NULL.
- Q49. Rename create_at to admission.
- Q50. Change the age datatype to INT.
- Q51. Set default phone number to 'N/A'.
- Q52. Remove the phone default.

Q53. Add a UNIQUE constraint to phone.

Q54. Remove that UNIQUE constraint.

Q55. Drop the is_active column.

Q56. Drop the country column. Do this after finishing the other ALTER TABLE exercises.

LEVEL 8 — UPDATE

Q57. Change the name of customer ID 1.

Q58. Change the age of customer ID 5 to 25.

Q59. Set the country of all customers to India.

Q60. Set is_active = FALSE for customers older than 35.

Q61. Set is_active = FALSE for customers whose name starts with A.

Q62. Update the phone number of customer ID 3.

LEVEL 9 — String Functions

Q63. Display all customer names in lowercase.

Q64. Display all customer names in uppercase.

Q65. Display name and name_length.

Q66. Find customers whose name length is greater than 4.

Q67. Display the first 3 characters of every name.

Q68. Display the last 3 characters of every name.

Q69. Extract characters 2–5 from each name.

Q70. Create a customer code using the first 3 letters of name + id. Example: Aarav → Aar1.

Q71. Create the format first 3 letters of name + id + age using CONCAT_WS().

Q72. Replace lowercase a with uppercase A in customer names.

LEVEL 10 — Math Functions

Q73. Calculate age + 5 for every customer.

Q74. Find ABS(-500).

Q75. Calculate the ceiling of 123.66.

Q76. Calculate the floor of 123.66.

Q77. Calculate the floor of -123.66.

Q78. Round 123.3034534 to 3 decimal places.

Q79. Truncate 123.4356457 to 2 decimal places.

Q80. Calculate 3^2 using POWER().

Q81. Calculate the square root of 121.

Q82. Find the remainder of 12 / 5 using MOD().

LEVEL 11 — Date & Time

- Q83. Display today's date.
- Q84. Display the current date and time.
- Q85. Display the year from create_at.
- Q86. Display the month from create_at.
- Q87. Display the week number from create_at.
- Q88. Display the day from create_at.
- Q89. Display the hour from create_at.
- Q90. Display the quarter from create_at.
- Q91. Find the age/difference between today's date and 2025-07-25.
- Q92. Truncate create_at to the year.
- Q93. Truncate create_at to the month.
- Q94. Truncate create_at to the week.
- Q95. Truncate create_at to the day.

LEVEL 12 — Real Data Analytics Questions

- Q96. Find the average age of customers whose names start with A.
- Q97. Find the number of customers aged between 20 and 30.
- Q98. Find the average age of customers aged between 20 and 30.
- Q99. Group customers by age and show only ages having more than one customer.
- Q100. Create an age group using CASE, then count customers in each age group.
- Q101. Display customers whose names start with A, ordered from oldest to youngest.
- Q102. Display the 3 oldest customers whose names contain a.
- Q103. Find the minimum and maximum age of customers whose name starts with A.
- Q104. Display name, age, uppercase_name, name_length, and age_group using UPPER(), LENGTH(), and CASE.
- Q105. Create a customer code using LEFT(name, 3), id, and age, separated using '- '.
- Q106. Display each customer's name, age, age + 5, and age * 2.
- Q107. Display customers who joined in a particular year using EXTRACT().
- Q108. Count how many customers joined in each month.
- Q109. Count how many customers joined in each year.
- Q110. Display name, create_at, year, month, and quarter.

FINAL CHALLENGE

- Challenge 1. Find the top 5 oldest customers.
- Challenge 2. Find the number of customers whose names start with A.
- Challenge 3. Find the average age of customers whose names start with A.

Challenge 4. Create name | age | age_group where <20 = Teen, 20–29 = Young, 30–39 = Adult, and 40+ = Senior.

Challenge 5. Count customers in every age_group.

Challenge 6. Find the age that occurs the most times. Hint: GROUP BY, COUNT, ORDER BY, LIMIT.

Challenge 7. Display each customer's name, first 3 characters, last 2 characters, name length, and uppercase name.

Challenge 8. Display the customer name and the year in which they were added.

Challenge 9. Find customers who were added after a particular date.

Challenge 10. Create a report containing name, age, age_group, name_length, uppercase_name, admission_year, and admission_month. Combine CASE + String Functions + Date Functions.